

Machine Learning GATE DA PYQs

GATE Data Science and Artificial Intelligence (GATE DA)

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Document compiled for GATE DA Preparation

Machine Learning (GATE DA)

1 GATE DA 2024

1.1 Supervised Learning: Linear Models & SVMs

Q1. Consider the dataset with six datapoints: $\{(x_1, y_1), (x_2, y_2), \dots, (x_6, y_6)\}$, where $x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $x_3 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$, $x_4 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$, $x_5 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$, $x_6 = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$ and the labels are given by $y_1 = y_2 = y_5 = 1$, and $y_3 = y_4 = y_6 = -1$. A hard margin linear support vector machine is trained on the above dataset. Which ONE of the following sets is a possible set of support vectors?

- (A) $\{x_1, x_2, x_5\}$
- (B) $\{x_3, x_4, x_5\}$
- (C) $\{x_4, x_5\}$
- (D) $\{x_1, x_2, x_3, x_4\}$

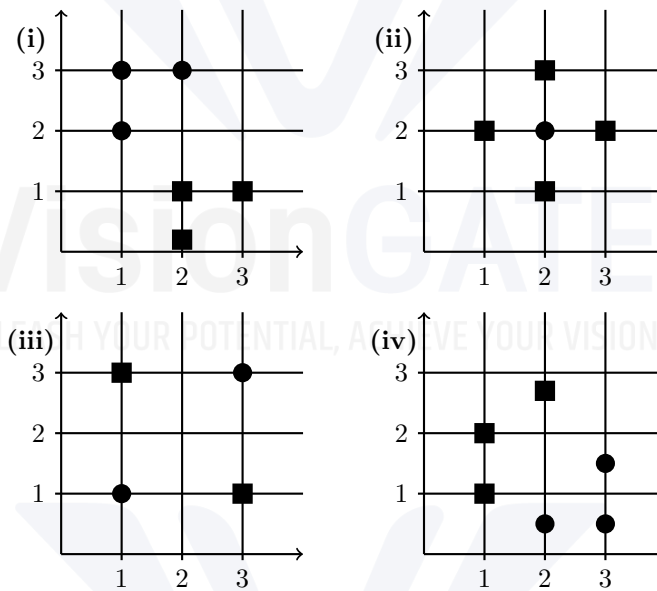
Q2. For any binary classification dataset, let $S_B \in \mathbb{R}^{d \times d}$ and $S_W \in \mathbb{R}^{d \times d}$ be the between-class and within-class scatter (covariance) matrices, respectively. The Fisher linear discriminant is defined by $u^* \in \mathbb{R}^d$, that maximizes

$$J(u) = \frac{u^T S_B u}{u^T S_W u}$$

If $\lambda = J(u^*)$, S_W is non-singular and $S_B \neq 0$, then (u^*, λ) must satisfy which ONE of the following equations? (Note: $\mathbb{R}$ denotes the set of real numbers.)

- (A) $S_W^{-1} S_B u^* = \lambda u^*$
- (B) $S_W u^* = \lambda S_B u^*$
- (C) $S_B S_W u^* = \lambda u^*$
- (D) $u^{*T} u^* = \lambda^2$

Q3. Consider the following figures representing datasets consisting of two-dimensional features with two classes denoted by circles and squares.



Which of the following is/are TRUE?

- (A) (i) is linearly separable.
- (B) (ii) is linearly separable.
- (C) (iii) is linearly separable.
- (D) (iv) is linearly separable.

1.2 Supervised Learning: Probabilistic Models, Trees & k-NN

Q4. Match the items in Column 1 with the items in Column 2 in the following table:

Column 1

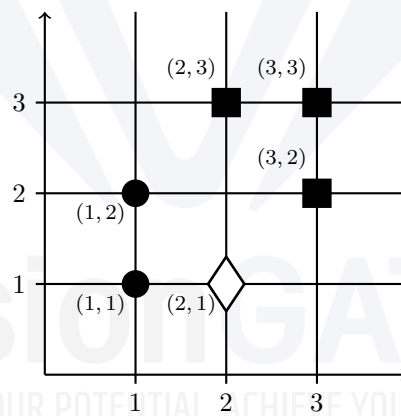
- (p) Principal Component Analysis
- (q) Naïve Bayes Classification
- (r) Logistic Regression

Column 2

- (i) Discriminative Model
- (ii) Dimensionality Reduction
- (iii) Generative Model

- (A) (p)-(iii), (q)-(i), (r)-(ii)
- (B) (p)-(ii), (q)-(i), (r)-(iii)
- (C) (p)-(ii), (q)-(iii), (r)-(i)
- (D) (p)-(iii), (q)-(ii), (r)-(i)

- Q5.** Given a dataset with K binary-valued attributes (where $K > 2$) for a two-class classification task, the number of parameters to be estimated for learning a naïve Bayes classifier is
- (A) $2K + 1$
 (B) $2^K + 1$
 (C) $2^{K+1} + 1$
 (D) $K^2 + 1$
- Q6.** Given the two-dimensional dataset consisting of 5 data points from two classes (circles and squares) and assume that the Euclidean distance is used to measure the distance between two points. The minimum odd value of k in k -nearest neighbor algorithm for which the diamond ($\diamond$) shaped data point is assigned the label square is _____.



- Q7.** Details of ten international cricket games between two teams "Green" and "Blue" are given in Table C. This table consists of matches played on different pitches, across formats along with their winners. The attribute Pitch can take one of two values: spin-friendly (represented as S) or pace-friendly (represented as F). The attribute Format can take one of two values: one-day match (represented as O) or test match (represented as T).
- A cricket organization would like to use the information given in Table C to develop a decision-tree model to predict outcomes of future games between these two teams. To develop such a model, the computed $\text{InformationGain}(C, \text{Pitch})$ with respect to the Target is _____ (rounded off to two decimal places).

Match Number	Pitch	Format	Winner (Target)
1	S	T	Green
2	S	T	Blue
3	F	O	Blue
4	S	O	Blue
5	F	T	Green
6	F	O	Blue
7	S	O	Green
8	F	T	Blue
9	F	O	Blue
10	S	O	Green

1.3 Unsupervised Learning: Clustering

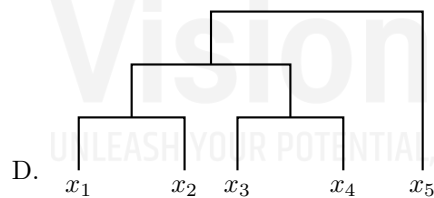
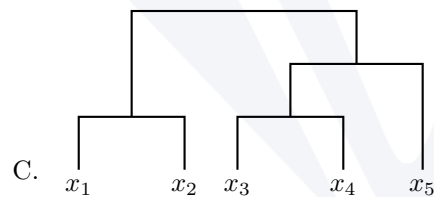
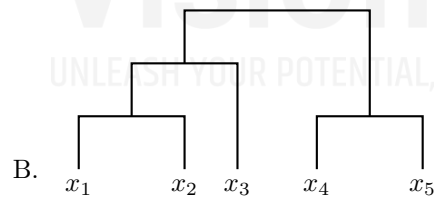
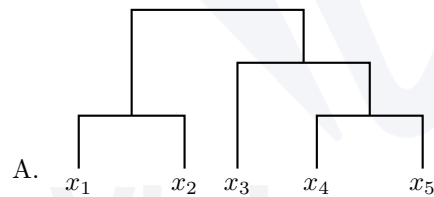
Q8. Euclidean distance based k -means clustering algorithm was run on a dataset of 100 points with $k = 3$. If the points $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ are both part of cluster 3, then which ONE of the following points is necessarily also part of cluster 3?

- (A) $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$
- (B) $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$
- (C) $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$
- (D) $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Q9. Consider the table below, where the $(i, j)^{\text{th}}$ element of the table is the distance between points x_i and x_j . Single linkage clustering is performed on data points x_1, x_2, x_3, x_4, x_5 .

	x_1	x_2	x_3	x_4	x_5
x_1	0	1	4	3	6
x_2	1	0	3	5	3
x_3	4	3	0	2	5
x_4	3	5	2	0	1
x_5	6	3	5	1	0

Which ONE of the following is the correct representation of the clusters produced?



1.4 Neural Networks & Deep Learning

Q10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function $f(x) = \frac{1}{1+e^{-x}}$. The value of the derivative of f at x where $f(x) = 0.4$ is _____. (rounded off to two decimal places).

(Note: $\mathbb{R}$ denotes the set of real numbers.)

Q11. Consider the two neural networks (NNs) shown in Figures 1 and 2, with *ReLU* activation ($\text{ReLU}(z) = \max\{0, z\}, \forall z \in \mathbb{R}$). The connections and their corresponding weights are shown in the Figures. The biases at every neuron are set to 0.

For what values of p, q, r in Figure 2 are the two NNs equivalent, when x_1, x_2, x_3 are positive?

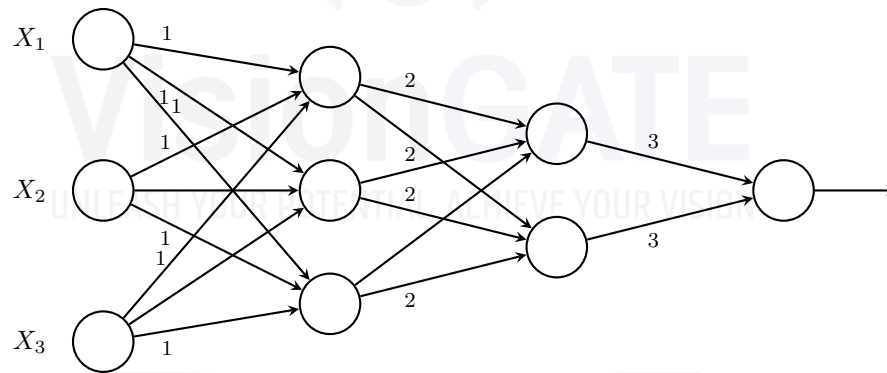


Figure 1

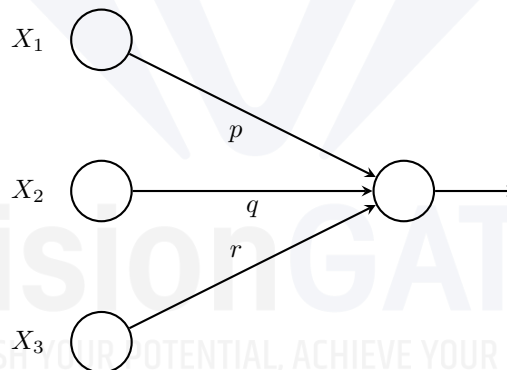


Figure 2

Note: $\mathbb{R}$ denotes the set of real numbers.

- (A) $p = 36, q = 24, r = 24$
- (B) $p = 24, q = 24, r = 36$
- (C) $p = 18, q = 36, r = 24$
- (D) $p = 36, q = 36, r = 36$

2 GATE DA 2025

2.1 Supervised Learning: Linear Models & SVM

Q12. Consider designing a linear classifier

$$y = \text{sign}(f(x; w, b)), \quad f(x; w, b) = w^\top x + b$$

on a dataset $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)\}$, $x_i \in \mathbb{R}^d$, $y_i \in \{+1, -1\}$, $i = 1, 2, \dots, N$. Recall that the sign function outputs +1 if the argument is positive, and -1 if the argument is non-positive. The parameters w and b are updated as per the following training algorithm:

$$w_{\text{new}} = w_{\text{old}} + y_n x_n, \quad b_{\text{new}} = b_{\text{old}} + y_n$$

whenever $\text{sign}(f(x_n; w_{\text{old}}, b_{\text{old}})) \neq y_n$. In other words, whenever the classifier wrongly predicts a sample (x_n, y_n) from the dataset, w_{old} gets updated to w_{new} , and likewise b_{old} gets updated to b_{new} . Consider the case $(x_n, +1), f(x_n; w_{\text{old}}, b_{\text{old}}) < 0$. Then

- (A) $f(x_n; w_{\text{new}}, b_{\text{new}}) > f(x_n; w_{\text{old}}, b_{\text{old}})$
- (B) $f(x_n; w_{\text{new}}, b_{\text{new}}) < f(x_n; w_{\text{old}}, b_{\text{old}})$
- (C) $f(x_n; w_{\text{new}}, b_{\text{new}}) = f(x_n; w_{\text{old}}, b_{\text{old}})$
- (D) $y_n f(x_n; w_{\text{old}}, b_{\text{old}}) > 1$

Q13. Given data $\{(-1, 1), (2, -5), (3, 5)\}$ of the form (x, y) , we fit a model $y = wx$ using linear least-squares regression. The optimal value of w is _____ (Round off to three decimal places).

Q14. Consider designing a linear binary classifier $f(x) = \text{sign}(w^\top x + b)$, $x \in \mathbb{R}^2$ on the following training data:

$$\text{Class-1: } \left\{ \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix} \right\}, \quad \text{Class-2: } \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$$

Hard-margin support vector machine (SVM) formulation is solved to obtain w and b . Which of the following options is/are correct?

- (A) $w = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$ and $b = 1$
- (B) The number of support vectors is 3
- (C) The margin is 2
- (D) Training accuracy is 98%

- Q15.** Consider a two-class problem in $\mathbb{R}^d$ with class labels red and green. Let μ_{red} and μ_{green} be the means of the two classes. Given test sample $x \in \mathbb{R}^d$, a classifier calculates the squared Euclidean distance (denoted by $\|\cdot\|_2$) between x and the means of the two classes and assigns the class label that the sample x is closest to. That is, the classifier computes

$$f(x) = \|\mu_{\text{red}} - x\|^2 - \|\mu_{\text{green}} - x\|^2$$

and assigns the label red to x if $f(x) < 0$, and green otherwise. Which of the following statements is/are correct?

- (A) The sample $x = 0$ is assigned the label green if $\|\mu_{\text{red}}\| < \|\mu_{\text{green}}\|$
- (B) f is a linear function of x
- (C) $f(x) = w^\top x + b$, where w and b are functions of μ_{red} and μ_{green}
- (D) f is a quadratic polynomial in x

2.2 Supervised Learning: Probabilistic Models

- Q16.** Which of the following statements is/are correct in a Bayesian network?

- (A) Variable elimination is an approximate inference algorithm
- (B) Gibbs sampling is an exact inference algorithm
- (C) Variable elimination is used to determine conditional probabilities
- (D) Rejection sampling is an approximate inference algorithm

- Q17.** The naïve Bayes classifier is used to solve a two-class classification problem with class-labels y_1, y_2 . Suppose the prior probabilities are $P(y_1) = \frac{1}{3}$ and $P(y_2) = \frac{2}{3}$. Assuming a discrete feature space with:

$$P(x | y_1) = \frac{3}{4} \text{ and } P(x | y_2) = \frac{1}{4}$$

for a specific feature vector x . The probability of misclassifying x is _____ (Round off to two decimal places).

2.3 Neural Networks & Deep Learning

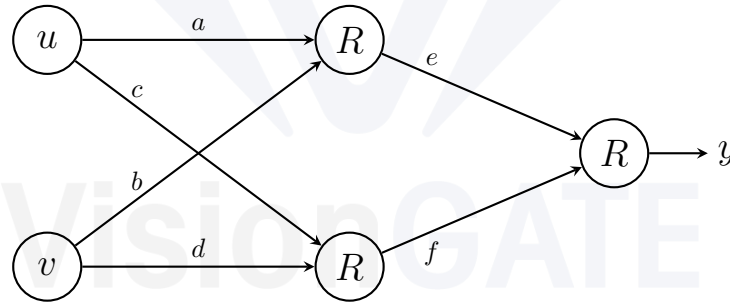
- Q18.** Consider the neural network shown in the figure with

inputs: u, v

weights: a, b, c, d, e, f

output: y

R denotes the ReLU function, $R(x) = \max(0, x)$.



Given $u = 2, v = 3$, and $a = 1, b = 1, c = 1, d = -1, e = 4, f = -1$, which one of the following is correct?

- (A) $\frac{\partial y}{\partial a} = 8, \frac{\partial y}{\partial f} = 0$
- (B) $\frac{\partial y}{\partial a} = 1, \frac{\partial y}{\partial f} = 0$
- (C) $\frac{\partial y}{\partial a} = 1, \frac{\partial y}{\partial f} = -1$
- (D) $\frac{\partial y}{\partial a} = 2, \frac{\partial y}{\partial f} = -1$

Q19. Which of the following statements is/are correct about the rectified linear unit (ReLU) activation function defined as $\text{ReLU}(x) = \max(x, 0)$, where $x \in \mathbb{R}$?

- (A) ReLU is continuous everywhere
- (B) ReLU is differentiable everywhere
- (C) ReLU is not differentiable at $x = 0$
- (D) $\text{ReLU}(x) = \text{ReLU}(ax)$, for all $a \in \mathbb{R}$

2.4 Unsupervised Learning: Clustering & PCA

Q20. Let C_1 and C_2 be two sets of objects. Let $D(x, y)$ be a measure of dissimilarity between two objects x and y . Consider the following definitions of dissimilarity between C_1 and C_2 :

$$\text{DIS-1}(C_1, C_2) = \max_{x \in C_1, y \in C_2} D(x, y)$$

$$\text{DIS-2}(C_1, C_2) = \min_{x \in C_1, y \in C_2} D(x, y)$$

Which of the following statements is/are correct?

- (A) Single Linkage Clustering uses DIS-1

- (B) Single Linkage Clustering uses DIS-2
- (C) Complete Linkage Clustering uses DIS-2
- (D) Complete Linkage Clustering uses DIS-1

Q21. Let $D = \{x^{(1)}, \dots, x^{(n)}\}$ be a dataset of n observations where each $x^{(i)} \in \mathbb{R}^{100}$. It is given that $\sum_{i=1}^n x^{(i)} = 0$. The covariance matrix computed from D has eigenvalues $\lambda_i = 100^{2-i}, 1 \leq i \leq 100$. Let $u \in \mathbb{R}^{100}$ be the direction of maximum variance with $u^\top u = 1$. The value of

$$\frac{1}{n} \sum_{i=1}^n (u^\top x^{(i)})^2 = \underline{\hspace{2cm}}$$

(Answer in integer).

3 GATE DA 2026

3.1 Machine Learning Fundamentals

- Q22.** In the following table, the Task column lists a few tasks related to machine learning. The Algorithm column lists a few algorithms. Each entry “t” from the Task column is to be matched with an appropriate entry “a” from the Algorithm column such that the task “t” can be solved using the algorithm “a”. Denote such a match as t:a.

Task	Algorithm
T1 – Clustering	A1 – Markov Chain Monte Carlo
T2 – Classification	A2 – K-Medoid
T3 – Sampling	A3 – Linear Discriminant Analysis
T4 – Feature Extraction	A4 – Naïve Bayes

Which of the following options is/are the correct matching(s)?

- (A) T1:A4, T2:A3, T3:A1, T4:A2
 (B) T1:A2, T2:A4, T3:A1, T4:A3
 (C) T1:A3, T2:A4, T3:A1, T4:A2
 (D) T1:A4, T2:A2, T3:A1, T4:A3

3.2 Unsupervised Learning: PCA & Clustering

- Q23.** For a classification problem, Principal Component Analysis (PCA) has been used to reduce the dimensionality of a feature space from 100 to 10. Which of the following options is true about the angle θ between the first and the tenth principal components?

- (A) $\theta = 0^\circ$
 (B) $\theta = 90^\circ$
 (C) $90^\circ < \theta \leq 180^\circ$
 (D) $0 < \theta < 90^\circ$

- Q24.** Let four points in three-dimensional space be:

$$P_1 : [2, 3, -1], \quad P_2 : [3, 1, 1], \quad P_3 : [5, -2, 3] \quad \text{and} \quad P_4 : [3, 3, 3]$$

Hierarchical Agglomerative Clustering is used to cluster the above points. If Manhattan Distance is used as the distance metric during clustering, which of the following options indicates the two points that will be merged first?

- (A) P_1, P_2
- (B) P_2, P_3
- (C) P_3, P_4
- (D) P_2, P_4

3.3 Supervised Learning: Regression & Regularization

Q25. Which of the following statements is true for Ridge Regression?

- (A) The regularizer in the objective function of Ridge Regression is used to guard against scenarios where the model works well for the test data, but poorly for the training data.
- (B) The regularizer of Ridge Regression uses L_1 norm.
- (C) Ridge Regression aims to reduce the number of parameters that have negative values.
- (D) The regularizer of Ridge Regression may increase the bias of the model, but it helps in reducing the variance in predictions.

Q26. Consider that Linear Ridge Regression is being used to learn a prediction function $y_{\text{pred}} = w^\top x$, where $w, x \in \mathbb{R}^2$ and Mean Absolute Error (MAE) is used to measure the prediction error. A weight of 0.20 is associated with the regularizer.

At an intermediate step of the training process, assume that the parameter $w = [-3.00, 4.00]^\top$. In the next step, for the input $x = [1.00, 2.00]^\top$, the predicted value of y is noted. Let the relation between $x = [x_1, x_2]^\top$ and the true value of y be $y_{\text{true}} = x_1 + x_2$.

The value of the overall regularized loss function for this instance is _____.
(Rounded off to two decimal places)

3.4 Model Evaluation & Metrics

Q27. Consider that you are training a classifier for a 10-class classification problem. Each input is represented as a 512-dimensional vector. There are 1000 samples, out of which first 100 will be used for testing. Let Leave-One-Out-Cross-Validation (LOOCV) be used for selection of the classifier model before testing. Which of the following options is the correct number of validation splits that will be generated?

- (A) 10
- (B) 512

- (C) 900
(D) 1000

- Q28.** Consider that 20 stories of Author X and 10 stories of Author Y were kept together without mentioning the names of the authors. A classifier was then asked to predict the author (X or Y) of each of these stories. Let, out of X's stories, 6 were classified as that of Y. On the other hand, out of Y's stories, 2 were classified as that of X. Considering X and Y as two classes, which of the following statements is/are true?
- (A) Accuracy of the classifier is $11/15$.
(B) Precision of Class X is higher than the Precision of Class Y.
(C) Recall of Class X is higher than the Recall of Class Y.
(D) Accuracy of the classifier is $14/15$.
-

3.5 Optimization & Gradient Descent

- Q29.** Consider that for a supervised learning task, the objective function being minimized is $f_w(x) = wx$, where $x \in \mathbb{R}$ is the input and $w \in \mathbb{R}$ is the parameter. Stochastic Gradient Descent with learning rate of 0.10 is used for parameter updates.
- Suppose that at the end of iteration i , the value of w becomes 10.00.
- Let $x = 10.00$ be the input for iteration $(i + 1)$.
- The value of w at the end of iteration $(i + 1)$ is _____. (Rounded off to two decimal places)
-

3.6 Neural Networks & Deep Learning

- Q30.** Consider a fully-connected feed-forward multi-layer perceptron. It has 30 neurons in the input layer, followed by two hidden layers and an output layer. The first hidden layer has 4 neurons and the second hidden layer has 3 neurons. The output layer has only one neuron. Assume that no bias parameters are used.
- The number of learnable parameters in the multi-layer perceptron is _____.
(Answer in integer)
-